

Graph Neural Networks for Defect Localization in Two-Layer CFRP from Ultrasonic NDT

Architecture study under a balanced noise-augmented recipe

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WCCM 2026 (draft June 7, 2026)

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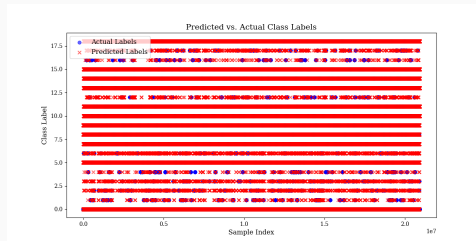
Motivation: NDT defect localization in layered CFRP

Carbon-fibre laminates around bolt holes accumulate **sub-surface delaminations** invisible to the eye.

Ultrasonic NDT yields a per-node response field on a **two-layer mesh** — but reading it by hand is slow and operator-dependent.

Goal

Predict, for every mesh node, *which of 19 defect classes* (incl. “defect-free”) it belongs to — directly from the mesh.



Predicted vs. true node labels on the hole mesh.

One specimen $\rightarrow$ one graph

- 13,942 nodes (two stacked layers around the hole)
- Node features: $[x, y, z, \text{DSPSS}]$
- Edges: FEM mesh connectivity (+ coordinate-difference edge features $[dx, dy, dz, \|\cdot\|]$)
- 19 classes; defect-free is class 0

Input = *noise-augmented difference* field (damaged – pristine), which isolates the defect signal.

The trap: extreme imbalance

Class-0 prior ≈ 0.998 . A model that predicts “defect-free” everywhere scores 99.8% accuracy and **macro-F1 ≈ 0** .

$\Rightarrow$ we report **macro-F1** and engineer the training distribution, not just the loss.

The recipe: balance the data, not (only) the loss

1. **Noise-augmented difference data** — robustness to acquisition noise.
2. **Defect-free negatives (NDF)** — 5000 genuine class-0 graphs added to *train only*.
3. **Balanced sampling** — `-defect_cap 5000 -ndf_count 5000` $\rightarrow$ 5000:5000.
4. Leakage control: group-purged val/test split by specimen.

Key finding

With the recipe in place, the *data distribution dominates*: a faithful reproduction reaches **macro-F1** ≈ 0.70 , vs. ≤ 0.08 without it.

A debugging detour: the silent collapse

Our refactored trainer **stalled at macro-F1 ≈ 0.16** across *every* architecture — while the legacy script reached 0.70.

Root cause: a default-on

`LogitAdjustLoss` ($\tau=1.5$). On a 0.998 class-0 prior it *over-boosts* minority classes $\Rightarrow$ mass over-prediction (high recall, precision ≈ 0).

Fix: `-no_logit_adjust` (Focal loss) $\Rightarrow$ full recovery.

Lesson

On extreme imbalance, logit adjustment and class balancing *compound*. Calibrate the loss to the *already-balanced* sampler, not the raw prior.

Corrected architecture sweep

Same recipe, 120 epochs, balanced 5000:5000, group-purged eval.

Architecture	macro-F1	note
MeshGraphNet	0.764	Encode-Process-Decode, edge update
GATv2	0.694	dynamic attention
GAT (Frontiers baseline)	0.674	prior published model (~ 0.61)
GraphSAGE	0.663	max-aggregation
PNA	0.613	multi-aggregator

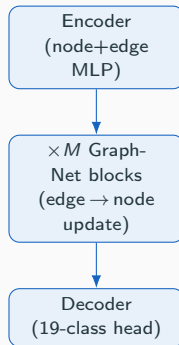
Takeaway

Under correct training, the **mesh-physics architecture (MeshGraphNet) wins** — beating the legacy GAT recipe (0.70) and the published baseline (0.61).

Why MeshGraphNet fits the problem

Encode–Process–Decode (Pfaff et al., 2021):

- node *and* edge embeddings updated at *every* message-passing step (residual);
- edge feature = coordinate difference $[dx, dy, dz, \text{dist}] \Rightarrow$ local geometry & anisotropy;
- matches FEM-mesh stress-propagation structure better than attention-only GNNs.



Novelty: four MeshGraphNet variants for layered NDT

Variant	Idea / why it could matter for CFRP
MeshGraphKAN	Fourier-KAN node encoder — learnable spectral basis for the <i>high-frequency stress-concentration</i> field (spectral-bias mitigation).
Hybrid-MGN	mesh edges + <i>front/back cross-layer “world” edges</i> — structurally separates the two laminate layers (no prior work).
BiStride-MGN	per-graph global-context broadcast — multiscale, long-range stress propagation (under-reaching).
MGN-Transformer	interleaved per-graph self-attention — global node interaction.

Hypothesis

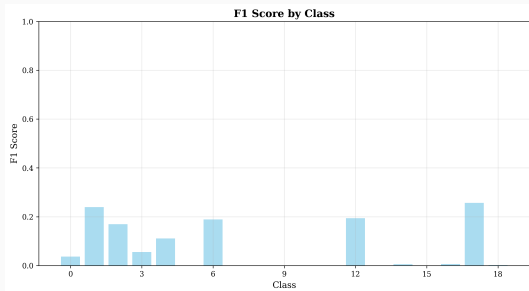
MeshGraphKAN × **stress concentration** and **Hybrid-MGN** × **two-layer separation** are the two novel angles most likely to push past 0.80.

Variant	macro-F1	vs. MeshGraphNet (0.764)
MeshGraphNet (baseline)	0.764	—
MeshGraphKAN	TBD	
Hybrid-MGN	TBD	
BiStride-MGN	TBD	
MGN-Transformer	TBD	

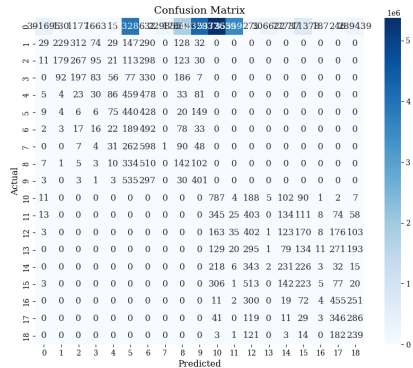
Target

Cross the **0.80** macro-F1 line with a physically-motivated variant.

Per-class performance & confusion



Per-class F1.



Confusion matrix.

Conclusions & outlook

- The **data recipe** (noise-aug difference + NDF negatives + balancing) is the dominant factor: 0.08 $\rightarrow$ 0.70.
- Under correct training, **MeshGraphNet** is the best architecture (**0.764**), beating the published GAT baseline (0.61).
- Two **novel layered-NDT variants** — Fourier-KAN and front/back Hybrid edges — target > 0.80 .

Future work

- **Application:** transfer to aerospace payload-fairing SHM (H3 / GNN-SHM).
- **Validation:** incorporate experimental (non-simulated) NDT data.

Thank you.

Questions?